

R1: SNR Axis and Measured Noise

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Over what signal-to-noise range is the inverse Identifiable, and where do the real sensors sit? Amir Dollie (DLLAMI004),
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SHARC buoy project Code: `runR1SNR.m`, `r1Boundary.m`, `plotR1SNR.m` Results: `Results/R1/R1_20261003-135701` (commit
a18e358)

Summary

Question. With the floe (p_0), layout (Level 2C, reference s26), sea (JONSWAP, $H_s = 0.05$ m, $\omega_p = 5$ rad/s, $\gamma_J = 3.3$), record ($N = 65536$) and estimator ($L = 2048$) fixed, sweep the accelerometer noise floor S_{nn} over four decades and locate every class boundary. **Outcome.** The inverse is Identifiable over a **window**, not below a single noise ceiling:

$$1.868 \times 10^{-7} \leq S_{nn} \leq 6.547 \times 10^{-6} \text{ (m/s}^2\text{)}^2 \text{ s/rad} \iff 110.4 \leq \text{ASD} \leq 653.8 \text{ } \mu\text{g}/\sqrt{\text{Hz}}.$$

The two ends fail for different reasons. At the noisy end precision runs out ($\sigma_{\ln \beta} = 1\%$). At the quiet end precision is excellent but the fixed finite-Welch bias becomes significant against it ($d_W = 0.5$). Four of the six noise cases lie inside the window: the fused estimate, the **LSM6DSV16X**, the G3 assumption and the **ICM-42688-P**. The LSM6DS3TR-C ($900 \text{ } \mu\text{g}/\sqrt{\text{Hz}}$) is precision-limited; the datasheet-grade case ($65 \text{ } \mu\text{g}/\sqrt{\text{Hz}}$) is bias-limited. No bin is rejected anywhere on the sweep, so neither boundary is a denominator artefact.

1 What the study does

One axis. The system is linear and the signal spectrum scales as H_s^2 , so for a fixed spectral shape only the ratio H_s^2/S_{nn} matters; one S_{nn} sweep is equivalently an H_s sweep (checked in the tester: scaling S_{nn} by 4 and H_s by 2 changes σ , d_W and F by less than 10^{-9}). Everything except S_{nn} is the E1 default, and the tester confirms the exact feature f_0 , the Jacobian J and the Welch feature bias b_W are identical at both ends of the axis; only $F = \sum_k J_k^T \Sigma_k^{-1} J_k$ changes.

Why there can be two boundaries. Lowering the noise raises F and so shrinks σ . The Welch feature bias b_W does not depend on the noise, but its significance $d_W = \sqrt{\delta \theta_W^T F \delta \theta_W}$ is measured against the shrinking uncertainty. A quiet enough sensor therefore turns a fixed estimator bias into a significant one.

Method (pre-registered in `cfg.R1`, clarification logged before the run). (1) 41 values of S_{nn} from 10^{-8} to 10^{-4} per rad/s, 0.1 decade apart (25.6 to 2555 $\mu\text{g}/\sqrt{\text{Hz}}$). (2) The six named noise cases evaluated exactly, not read off the grid. (3) Every class change between neighbouring grid points refined by bisection in $\log_{10} S_{nn}$ on the metric that crosses its threshold, to 10^{-3} decade (`r1Boundary.m`; 9 evaluations each). (4) Each boundary also expressed as the H_s at which the LSM6DSV16X would have that SNR, $H_{s,\text{eq}} = 0.05 \sqrt{S_{\text{LSM}}/S_{nn}}$ m. ASD in $\mu\text{g}/\sqrt{\text{Hz}}$ is $\sqrt{2\pi S_{nn}}/9.81 \times 10^{-6}$.

2 Results

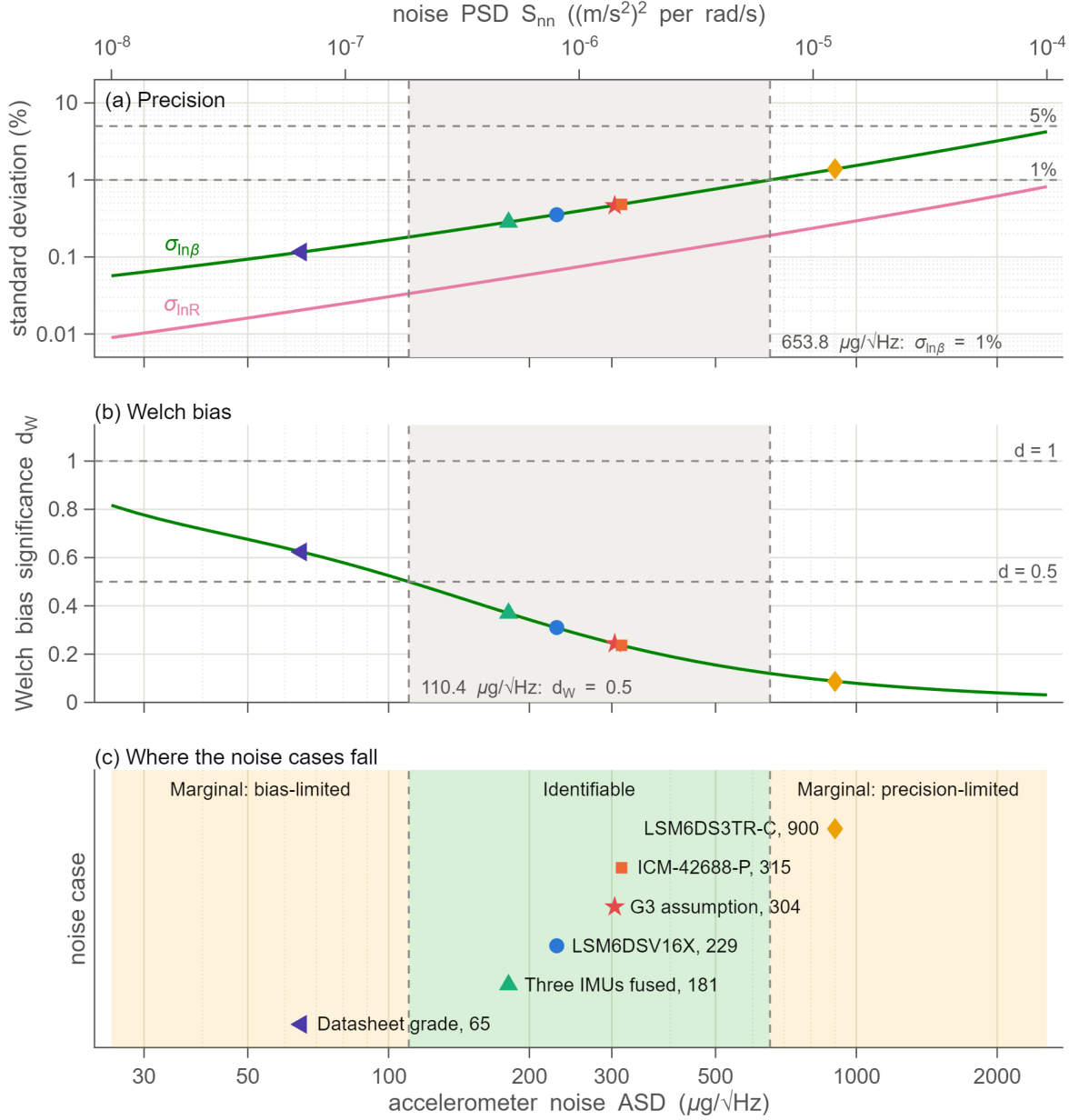


Figure 1. The SNR window for p_0 , Level 2C, $N = 65536$, $L = 2048$. Bottom axis: noise ASD; top axis: the same noise as a PSD per rad/s. (a) $\sigma_{\ln\beta}$ and $\sigma_{\ln R}$ with the 1% and 5% limits. (b) Welch bias significance with the $d = 0.5$ and 1 limits. (c) The six noise cases (exact evaluations) over the class band of the sweep. Shaded in (a) and (b): the Identifiable interval between the refined boundaries (dashed verticals), each labelled in the panel whose criterion binds.

2.1 Three regimes

Regime	ASD ($\mu g/\sqrt{\text{Hz}}$)	S_{nn} (per rad/s)	Class	Binding metric
Bias-limited	25.6 to 110.4	10^{-8} to 1.868×10^{-7}	Marginal	d_W 0.817 down to 0.5
Window	110.4 to 653.8	1.868×10^{-7} to 6.547×10^{-6}	Identifiable	both met
Precision-limited	653.8 to 2555	6.547×10^{-6} to 10^{-4}	Marginal	$\sigma_{\ln\beta}$ 1% up to 4.23%

Window width: a factor 35 in PSD, 5.9 in ASD (1.54 decades). Bad conditioning never binds: $\kappa(F)$ stays between 96 and 125. Not identifiable is not reached on the sweep (it would need $d_W > 1$ or $\sigma > 5\%$).

$\sigma_{\ln R}$ follows $\sigma_{\ln\beta}$ at about one fifth of its value (0.009 to 0.818%) and never binds. In the noisy half $\sigma_{\ln\beta}$ grows as $\sqrt{S_{nn}}$ (a factor 3.13 from 10⁻⁶ to 10⁻⁵, against $\sqrt{10} = 3.16$) and d_W falls in proportion. Towards the quiet end the gain flattens (a factor 2.4 per decade from 10⁻⁸ to 10⁻⁷): each further decade of sensor quality buys less precision while the bias stays fixed.

2.2 Where the noise cases sit

Noise case	ASD ($\mu g/\sqrt{\text{Hz}}$)	PSD/G3	ASD/G3	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	Class	H_s window (m)
Datasheet grade	65.1	0.046	0.214	0.116%	0.020%	0.624	Marginal (bias)	0.005 to 0.030
Three IMUs fused	180.7	0.353	0.594	0.284%	0.053%	0.369	Identifiable	0.014 to 0.082
LSM6DSV16X	228.5	0.565	0.752	0.355%	0.067%	0.309	Identifiable	0.018 to 0.104
G3 assumption	304.1	1.000	1.000	0.468%	0.089%	0.244	Identifiable	0.023 to 0.138
ICM-42688-P	315.0	1.073	1.036	0.485%	0.092%	0.236	Identifiable	0.024 to 0.143
LSM6DS3TR-C	899.8	8.757	2.959	1.382%	0.264%	0.088	Marginal (precision)	0.069 to 0.407

H_s window: the significant wave heights of this spectral shape for which that sensor lies inside the window, $0.05\sqrt{S/S_{\text{hi}}}$ to $0.05\sqrt{S/S_{\text{lo}}}$ m. The lower H_s limit is the precision boundary, the upper one the bias boundary.

The G3 assumption was realistic. The ICM-42688-P sits almost exactly on it (1.07 in PSD, 1.04 in ASD) and the LSM6DSV16X is 25% quieter in ASD. The earlier work that used G3 as the stress noise level did not rest on an unrealistic noise scale.

0% invalid bins everywhere. The estimator's denominator rule rejects no bin at any S_{nn} from 10^{-8} to 10^{-4} (all 84 bins used), so both boundaries come from the two mechanisms the study was designed to expose, Fisher precision and Welch-bias significance, and not from bins dropping out.

3 Interpretation

1. **The inverse has an SNR window at fixed L .** Below $110.4 \mu g/\sqrt{\text{Hz}}$ it is estimator-bias limited; above $653.8 \mu g/\sqrt{\text{Hz}}$ it is noise limited. This is the headline result of R1.
2. **A quieter sensor is not worse.** The datasheet-grade case carries more information than any other case ($\sigma_{\ln \beta} = 0.116\%$); it is Marginal only because the $L = 2048$ Welch bias becomes significant against that small uncertainty. The fix is a finer spectral resolution (larger L ; D2 and R2 show the bias falls by about a factor 4 per doubling of L), not added noise. The quiet boundary is a property of the estimator setting, not a requirement on the hardware.
3. **The same in H_s .** For the LSM6DSV16X the basin sea should have $0.018 \lesssim H_s \lesssim 0.104$ m at this spectral shape. The lower limit is precision; the upper limit is bias significance. Larger waves do not carry less information; with $L = 2048$ fixed, the Welch bias becomes the limiting error once the random error is small enough. The reference $H_s = 0.05$ m lies inside it, a little above its geometric centre (0.043 m).
4. **Consistent with D2.** D2 found the datasheet-grade sensor Marginal at 109 min because d_W grows relative to the uncertainty as the record lengthens. R1 shows the same along the SNR axis: more statistical information does not mean a smaller standardised estimator bias. Once the experiment is precise enough, the estimator, not the sensor, sets the limit.
5. **Both candidate sensors are adequate with margin.** The LSM6DSV16X is a factor 2.9 in ASD inside the noisy boundary and 2.1 inside the quiet one; the ICM-42688-P a factor 2.1 and 2.9. The LSM6DS3TR-C is not adequate at this sea and record ($\sigma_{\ln \beta} = 1.38\%$); it would need $H_s \geq 0.069$ m or a longer record.
6. **Fusing three on-board IMUs helps precision** ($\sigma_{\ln \beta}$ 0.284% against 0.355% for the LSM6DSV16X alone) and stays inside the window, but it moves the operating point towards the quiet boundary ($d_W = 0.369$). At the planned L the gain is real but not decisive.

Limitations. One floe and one layout (p_0 , Level 2C); A1 showed the precision depends on the floe, so the window will move across the domain. One spectral shape (R2 covers shape; the H_s equivalence holds only at fixed shape). One record length and one L : the quiet boundary is specific to $L = 2048$ and $N = 65536$ and would move with either. White, known noise (R3 covers a misspecified floor). Analytic expectations, not realised records (M1 tests the selected case with Monte Carlo records). Model truncation bias (A1) is outside this within-model analysis.

Requirement for the experiment and the thesis

For p_0 , Level 2C, $N = 65536$ and $L = 2048$, the inverse is Identifiable for noise ASD between 110.4 and $653.8 \mu g/\sqrt{\text{Hz}}$ (equivalently, for the LSM6DSV16X, $0.018 \leq H_s \leq 0.104$ m at the reference spectral shape). Both candidate sensors, the LSM6DSV16X ($229 \mu g/\sqrt{\text{Hz}}$) and the ICM-42688-P ($315 \mu g/\sqrt{\text{Hz}}$), meet this with a margin of about 2 either side; the LSM6DS3TR-C does not. The quiet-end boundary is an estimator limit: if a quieter sensor or larger waves are used, increase L rather than accept the bias. **M1 hand-off:** p_0 at LSM6DSV16X noise, an interior point of the window.

Outputs (Results/R1/): R1.20261003_135701.log/.mat/_summary.txt; R1_sweep.csv, R1_sensor_points.csv, R1_boundaries.csv; figures/R1_fig1_snr_window (.pdf/.png).